

```
from typing import List
```

```
1 def factorize(n: int) -> List[int]:
2     """ Return list of prime factors of given
3     integer in the order from smallest to largest.
4     Each of the factors should be listed number of
5     times corresponding to how many times it
6     appears in factorization.
7     Input number should be equal to the product of
8     all factors
9     >>> factorize(8)
10         [2, 2, 2]
11     >>> factorize(25)
12         [5, 5]
13     >>> factorize(70)
14         [2, 5, 7]
15     """
```

Prompt

```
METADATA = {
    'author': 'jt',
    'dataset': 'test'
}
```

```
1 def check(candidate):
2     assert candidate(2) == [2]
3     assert candidate(4) == [2, 2]
4     assert candidate(8) == [2, 2, 2]
5     assert candidate(3 * 19) == [3, 19]
```

Test case

GitHub Copilot Suggestions
- Suggestion 1

```
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13     >>> factorize(70)
14         [2, 5, 7]
15     """
16     result = []
17     i = 2
18     while i <= n:
19         if n % i == 0:
20             result.append(i)
21             n //= i
22         else:
23             i += 1
24     return result
```

Generated code